

Reproductive patterns and inflammation: An evolutionary mismatch perspective

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BACKGROUND

Women in modern industrialised societies experience far more menstrual cycles than women in ancestral environments. Earlier menarche, fewer pregnancies, shorter breastfeeding durations, and widespread hormonal contraceptive use reduce time spent pregnant or lactationally amenorrheic. This creates a reproductive evolutionary mismatch, increasing cumulative ovulatory cycles and lifetime exposure to oestrogen^{1, 2}. Ovulation is an inflammatory process, and repeated cycles are associated with heightened systemic inflammation³. Related pathways have been linked to common female-biased conditions such as endometriosis, uterine fibroids, ovarian cysts, and irritable bowel syndrome (IBS)¹⁻⁵. These overall linking theories have been schematised in figure 1.

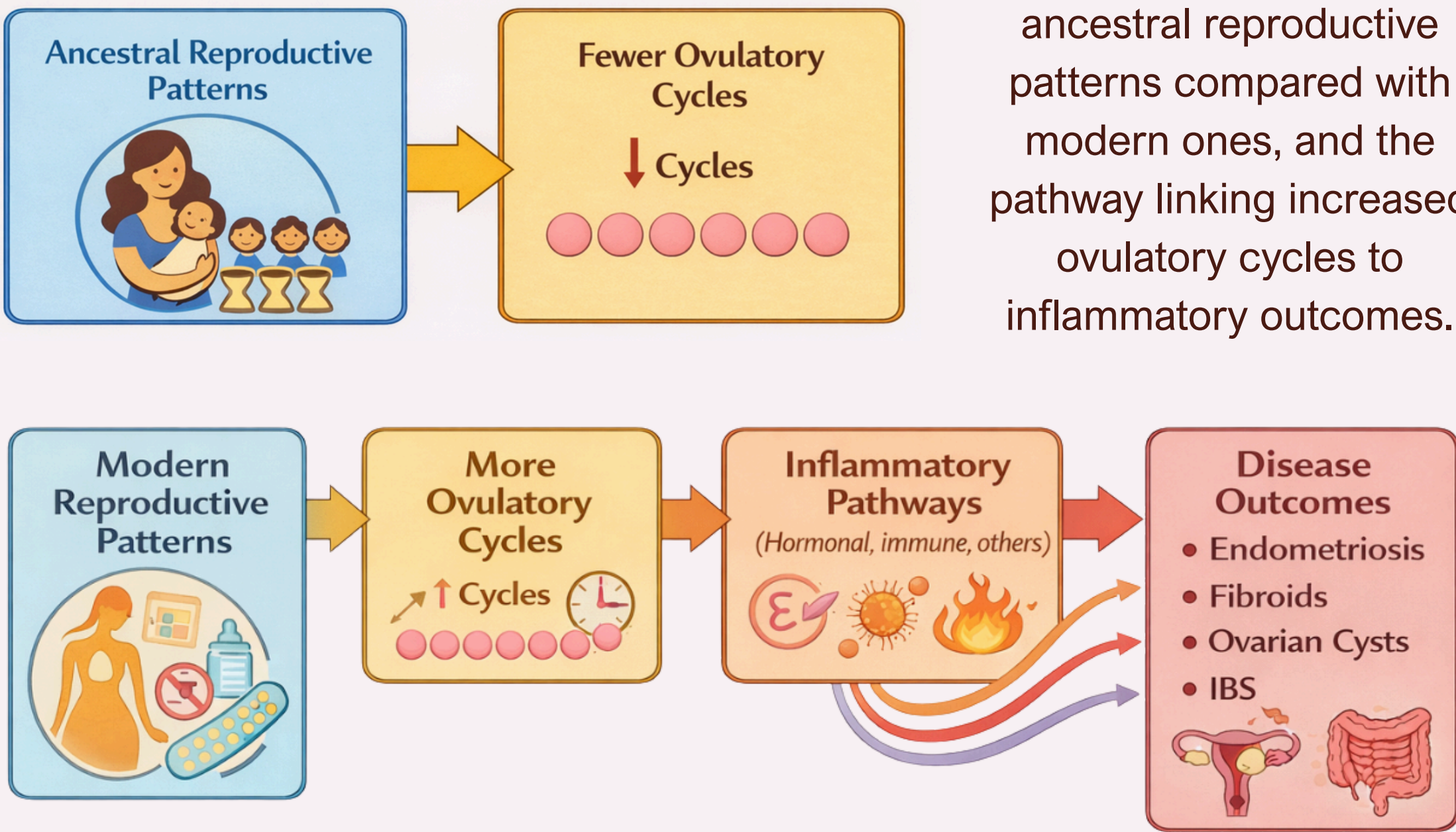


Figure 1: Hypothesised ancestral reproductive patterns compared with modern ones, and the pathway linking increased ovulatory cycles to inflammatory outcomes.

RESEARCH QUESTIONS

- Do modern reproductive patterns predict inflammatory reproductive conditions and IBS?
- Does IBS have similar associations with mismatch as gynaecological conditions in this cohort?
- Does hormonal contraceptive use differ from other mismatch factors in its association with disease risk?

DATA

UCL National Survey of Health and Development (NSHD)1946 Cohort - Women's Health Questionnaire.

- Predictors: age at menarche, parity, breastfeeding duration, contraceptive use
- Outcomes (binary): endometriosis, uterine fibroids, ovarian cysts, IBS.

HYPOTHESIS

I hypothesise that women with:

- earlier menarche,
- and/or fewer children,
- and/or having given birth later in life,
- and/or who have spent less time breastfeeding

would have increased incidence of outcomes in post-reproductive life.

Contraceptive use may either increase odds of inflammatory disease, or act as a protective mechanism by reducing ovulatory cycles.

PROPOSED ANALYSIS

Multivariable logistic regression in R:

- predictors will be entered simultaneously in a multivariable model to assess their relative contributions to each outcome. Some preliminary descriptive statistics in figure 2 already describe cohort characteristics.

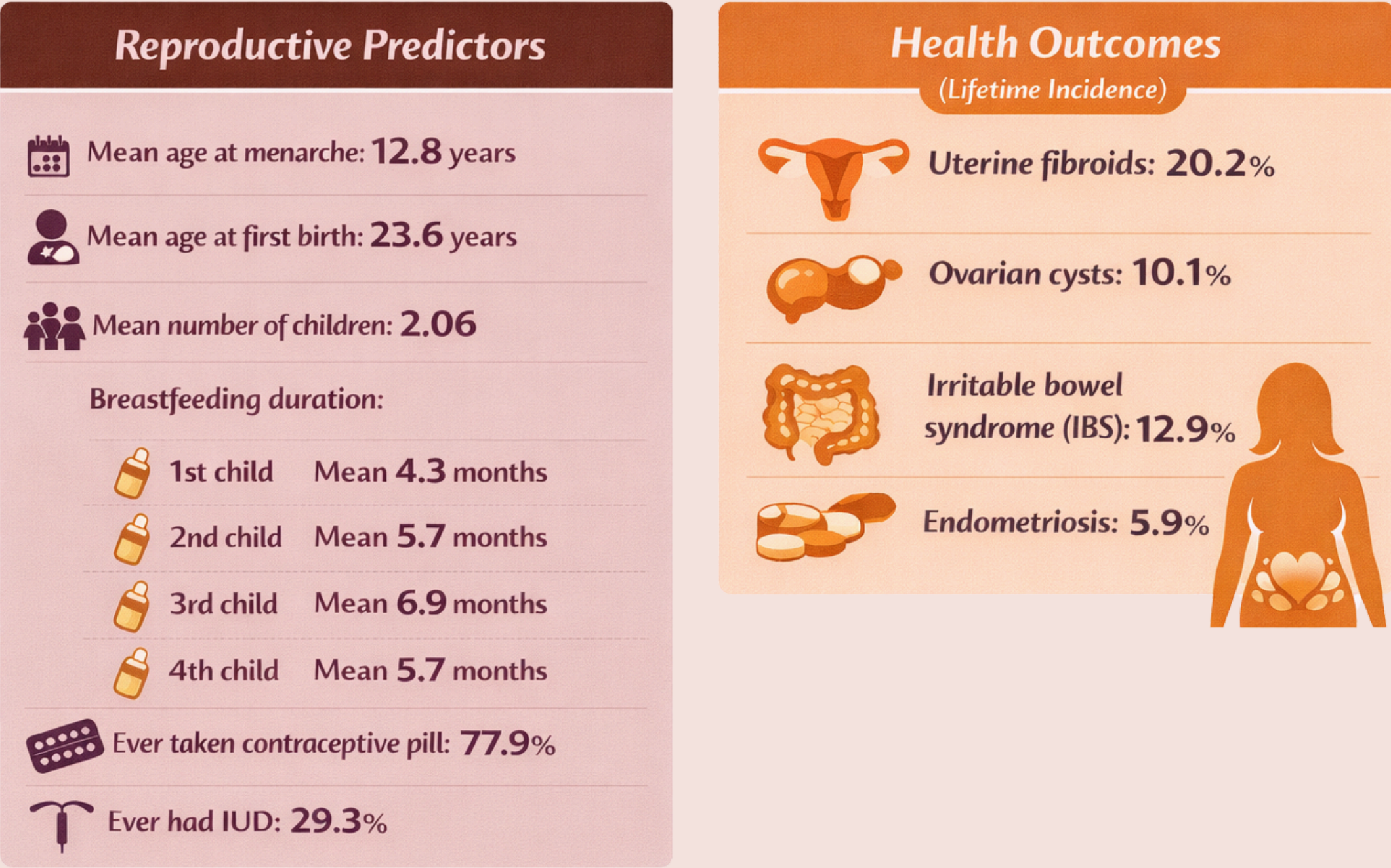
Control for:

- socioeconomic status, smoking status
- heavy/painful periods, hysterectomy (contraceptive indication bias)
- make sure disease outcome is after reproductively active

Supplementary:

- time analysis between age at birth and age at onset of disease.

Figure 2: Summary statistics calculated so far in R studio for NSHD data.



STRENGTHS AND WEAKNESSES

Strengths: Prospective, well-characterised cohort; integration of reproductive and gastrointestinal outcomes

Limitations: Self-reported diagnoses; WEIRD population limits generalisability; observational design cannot show specific cause and effect.

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